

Reviewer Pack — Minimal Diophantine Exponent Bounds Circuit Complexity of Ra...

Entry 11ad4596b4aa · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Diophantine Exponent Bounds Circuit Complexity of Random k-CNF

Entry ID: 11ad4596b4aa **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-30 12:45:10 UTC

1. Conjecture

Field A (mathematical branch): Number Theory (Diophantine Exponents) **Field B** (complexity object): Boolean Function Complexity: Circuit Complexity

Statement:

For a random k-CNF formula with n variables, the minimal diophantine exponent $\delta(\varphi)$ required to solve φ is at most $O(k^{3n}(1/4))$ where k is the clause length.

Rationale (proposer's reasoning):

Number theory, particularly the study of Diophantine exponents, provides a rich set of tools for understanding the complexity of computational problems. If this conjecture holds, it would suggest that the difficulty of satisfiability in random k-CNF formulas is closely tied to diophantine equations, potentially revealing new insights into circuit complexity.

Taxonomy category: cg_kw_andreev (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 134df9e2e585b1de

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the maximum value of $\delta(\varphi)$ across all generated k-CNF formulas with n variables and clause length $k \leq 40$ is less than or equal to $O(k^{3n}(1/4))$. It is falsified if this maximum exceeds $O(k^{3n}(1/4))$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 3 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Diophantine Exponent Bounds" AND "Circuit Complexity" AND k-CNF" - "Random k-CNF" AND "Minimal Diophantine Exponent" AND circuit complexity" - "Number Theory" AND diophantine exponent AND Boolean function complexity

Top relevant hits considered: - [http://arxiv.org/abs/1609.06986v1] More on Diophantine sextuples - [http://arxiv.org/abs/2406.18700v4] Structure of sparse Boolean functions over Abelian groups, and its application to testing - [http://arxiv.org/abs/2103.08351v3] Initial nonrepetitive complexity of regular episturmian words and their Diophantine exponents

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_kcnf(n, k):
        cnf = []
        for _ in range(k*n):
            clause = [random.randint(1, n), -random.randint(1, n)]
            if len(set(clause)) == 2:
                cnf.append(clause)
        return cnf

    def diophantine_exponent(cnf):
        # Placeholder implementation of the diophantine exponent calculation
        # This is a dummy function that returns a constant value for demonstration purposes
        return 1.0

    max_delta = 0.0
    instances_tested = 0
    n_max = 0

    for n in [5, 10, 15, 20, 30, 40]:
        for _ in range(5): # Ensure at least 30 instances per seed
            cnf = generate_kcnf(n, 4) # Assuming k=4 for simplicity
            delta = diophantine_exponent(cnf)
            max_delta = max(max_delta, delta)
            instances_tested += 1
```


8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test code uses a placeholder implementation for the diophantine exponent calculation, which returns a constant value. This does not provide any empirical evidence of the actual complexity of solving the k-CNF formula and thus cannot support or falsify the conjecture.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code uses a placeholder implementation for the diophantine exponent calculation that returns a constant value, which does not provide empirical evidence of the actual complexity of solving k-CNF formulas. Therefore, it cannot support or falsify the conjecture.
| next: Develop a proper implementation to calculate the diophantine exponent for k-CNF formulas and conduct a new test to properly evaluate the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	26915
2	propose	ollama_remote	glm4:latest	0	0	22713
3	preregistration	ollama_remote	glm4:latest	0	0	22943
4	novelty	ollama_remote	glm4:latest	0	0	8571
5	novelty	ollama_remote	glm4:latest	0	0	11639
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13761
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10271
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14233
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8287
10	critic	ollama_remote	glm4:latest	0	0	17031
11	judge	ollama_remote	glm4:latest	0	0	9395

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 165760 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in research/audit/11ad4596b4aa.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/11ad4596b4aa.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/11ad4596b4aa.tar.gz (if generated)